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Pose-oriented scene-adaptive matching for abnormal event detection

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ABSTRACT

For intelligent surveillance systems, abnormal event detection automatically analyses surveillance video sequences and detects abnormal objects or unusual human actions at the frame level. Due to the lack of labelled data, most approaches are semi-supervised based on reconstruction or prediction methods. However, these methods may not generalize well to unseen scene contexts. To address this issue, we present a novel self and mutual scene-adaptive matching method for abnormal event detection. In the framework, we propose synergistic pose estimation and object detection, which effectively integrates human pose and object detection information to improve pose estimation accuracy. Then, the poses are resized to reduce the spatial distance between the source and target domains. The improved pose sequences are further fed into a spatio-temporal graph convolutional network to extract the geometric features. Finally, the features are embedded in a clustering layer to classify action types and compute normality scores. The training data is taken from the training part of common video anomaly detection datasets: UCSD PED1 & PED2, CHUK Avenue, and ShanghaiTech Campus. The proposed framework is evaluated on video sequences with unseen scene contexts in the UCSD PED2 and ShanghaiTech Campus datasets. The detection accuracy and efficiency are also evaluated in detail, and the proposed method for abnormal event detection achieves the highest AUC performance, 84.6%, on the ShanghaiTech Campus dataset and relatively high AUC performance, 96.9% and 74.8%, on UCSD PED2 & PED1 datasets. Compared with other state-of-the-art works, the performance analysis and results confirm the robustness and effectiveness of our proposed framework for cross-scene abnormal event detection.

1. Introduction

Abnormal event detection (AED), the practice of identifying, analysing, and classifying anomalies at the frame level in video sequences, is an area of study experiencing significant growth [1–3]. Its applications span diverse fields, including intelligent surveillance systems, healthcare, robotics, and human–computer interfaces [4–7]. The increase in relevance and focus on AED can be attributed to the growing integration of surveillance technologies in contemporary societal infrastructures.

Presently, most AED techniques, such as [8–11], struggle with the scarcity of labelled data, leading to an overreliance on reconstruction or prediction models in the semi-supervised learning context. While reconstruction methods have seen success in modelling normal events, and prediction methods have accurately determined normal event probability distributions, these methods share a common shortcoming. They often fail to generalize across differing scenes or contexts, leading to limitations in their applicability and effectiveness [2,12–14].

Cross-scene AED represents a critical frontier in the field, promising to reduce computational requirements in intelligent surveillance systems significantly. Current efforts in this direction, such as the evaluation explored in [15], have proven challenging in that difficulties stem from complexities introduced by diverse background information and varying definitions of abnormal events across different datasets. Our previous work [16] has proven that background information is beneficial for same-scene abnormal recent detection since in many video streams, especially in surveillance, the background remains relatively static while the action takes place in the foreground. By modelling and subtracting the background, the anomalies or events in the foreground can be highlighted and analysed more efficiently. Meanwhile, the background provides the context information. Anomalies cannot always be defined strictly by motion or appearance alone. Sometimes, the context provided by the background is vital. For instance, a person running might be normal in the park but could be considered abnormal on the lawn or the driveway. Fig. 2 shows three different image-level inputs for AED, original, motion, and object images. According to the results of same-scene AED, the original images can achieve the best detection accuracy and the highest computational cost. While object

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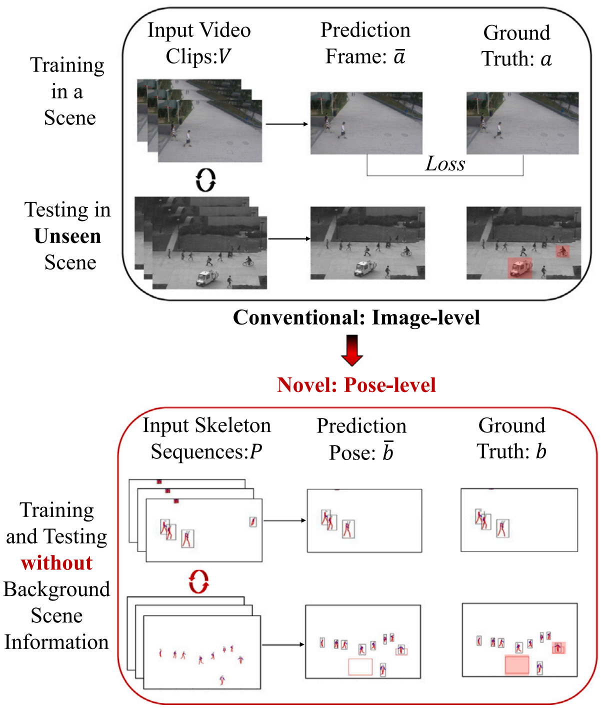

Fig. 1. Problem review. Scene-adaptive works for AED are trained in the source domain and tested in the target domain. Compared with the image-level works, our pose-oriented AED is relatively more explainable and efficient.

images can reduce computational costs at a low level, their detection performance is much worse than that of original and motion images. However, background information may be helpful for detecting same-scene anomalies, but it is a burden to detect cross-scene anomalies after cross-scene AED evaluation. This inconsistency limits the ability to train a universally applicable model, leading to unsatisfactory performance in cross-scene contexts.

Recognizing the limitations of existing methods, this paper introduces a novel approach centred on pose-oriented AED. The advantage of focusing on human poses lies in their potential as robust, scene-agnostic features suitable for cross-scene learning. Unlike more superficial or context-specific features, human poses present a more consistent and interpretable basis for AED.

The proposed scene-adaptive framework leverages synergistic pose estimation and object detection. Effectively integrating these components enhances pose information accuracy. The method further utilizes spatial and temporal graph convolutional networks to learn spatio-temporal feature representations, culminating in a powerful new approach to AED. The main contributions of this work encompass the following:

- Introducing a novel pose-oriented scene-adaptive framework for AED in surveillance videos and addressing a recognized gap in current methodologies, as illustrated in Fig. 1.
- Developing a new method by combining pose estimation and object detection and paving the way for more accurate cross-scene adaptive AED.
- Demonstrating the effectiveness of the proposed framework through rigorous validation of public AED datasets and substantiating its superior performance relative to existing state-of-the-art methods.

In summary, this paper not only identifies a critical challenge within the field of AED but also presents a pioneering approach to overcome it. Preliminary aspects of this work, primarily addressing cross-scene AED through the fusion of pose and class information, have previously been presented in [17]. The current pose-oriented methods mark a substantial advancement in the field, providing a pathway towards more efficient, accurate, and widely applicable AED solutions. The implications of this work could extend well beyond the surveillance field, contributing to broader advances in machine learning and artificial intelligence.

2. Related work

Deep learning-based methods are widely used for AED. The classic paradigm involves training on normal videos to establish the probability distribution of normal samples and analysing errors to distinguish normal and abnormal samples in test videos. Based on these errors, two main types of methods exist for AED: reconstruction-based and predictive methods. Recently, researchers have identified cross-scene AED as a new research direction, aiming to build a unified system capable of detecting different anomalies across various scenarios. Object-centric methods and human pose estimation algorithms are utilized in AED, which are novel approaches for cross-scene AED. Pose graphs are unaffected by the background, making them more effective for behaviour analysis.

2.1. Deep learning techniques for AED

Reconstruction-based methods are prevalent in recent AED work [7,16,18–20]. Image data are trained in autoencoder models to generate latent vectors of normal samples. Testing video sequences similar to the training data will be reconstructed with low errors. In contrast, strange actions or rare events that deviate from normal data are expected to be reconstructed with higher error. Refs. [7,16,20] provide additional approaches where attention-based methods are added to reconstruction-based models for the original data. These methods split the input images into foreground and background images, focusing only on the motion area. Despite their seeming effectiveness, reconstruction-based methods also have some limitations. Firstly, these methods are not sensitive to cluttered or crowded background scenes. While they can detect deviations from normal behaviour, subtler abnormalities may not result in significant reconstruction errors, leading to missed detections. Moreover, these methods often fail to effectively discriminate between actual abnormal events and outliers or noise in the data, leading to false positives. Finally, their lack of robust cross-scene generalization capabilities presents a considerable challenge in real-world applications.

Predictive methods constitute another crucial branch of AED [21–23]. These techniques learn the probability distribution of normal events by predicting future frames based on historical data. Like reconstruction-based methods, they work under the premise that abnormal events will disrupt the normal event pattern, causing the predictive model to generate high prediction errors. Typically, these methods are built upon recurrent neural networks or other temporal models capable of understanding sequential data. By predicting what comes next in a video sequence, these models can ascertain when a deviation from the normal pattern occurs and identify abnormal events. However, like their reconstruction-based counterparts, predictive methods have limitations. One of the most critical drawbacks is distinguishing between actual abnormal events and unexpected but normal variations in data, leading to a high false-positive rate [23]. In addition, they might fail to detect subtle abnormalities that do not significantly disrupt the overall pattern of normal events. Another challenge arises in defining the degree of error that separates normal from abnormal events, which might vary significantly across different scenarios. Furthermore, these models lack robust cross-scene generalization capabilities compared to reconstruction-based methods, limiting their utility in diverse real-world settings.

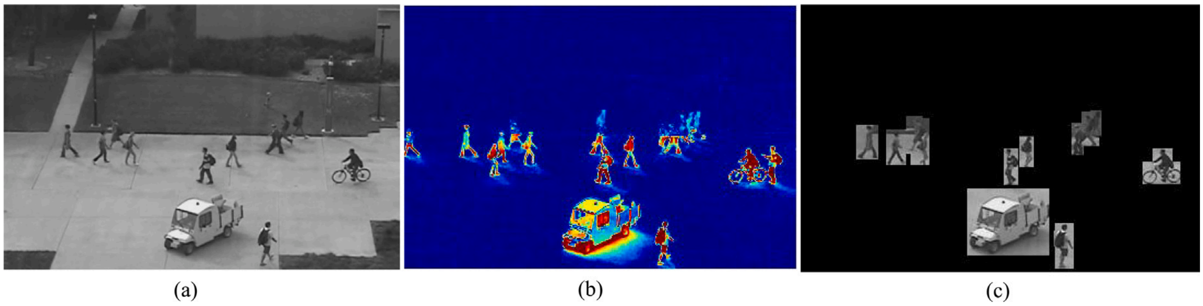

Fig. 2. Three different inputs for image-level AED works. (a) is the original image. (b) is the motion image after motion estimation. (c) is the object image after object detection and segmentation.

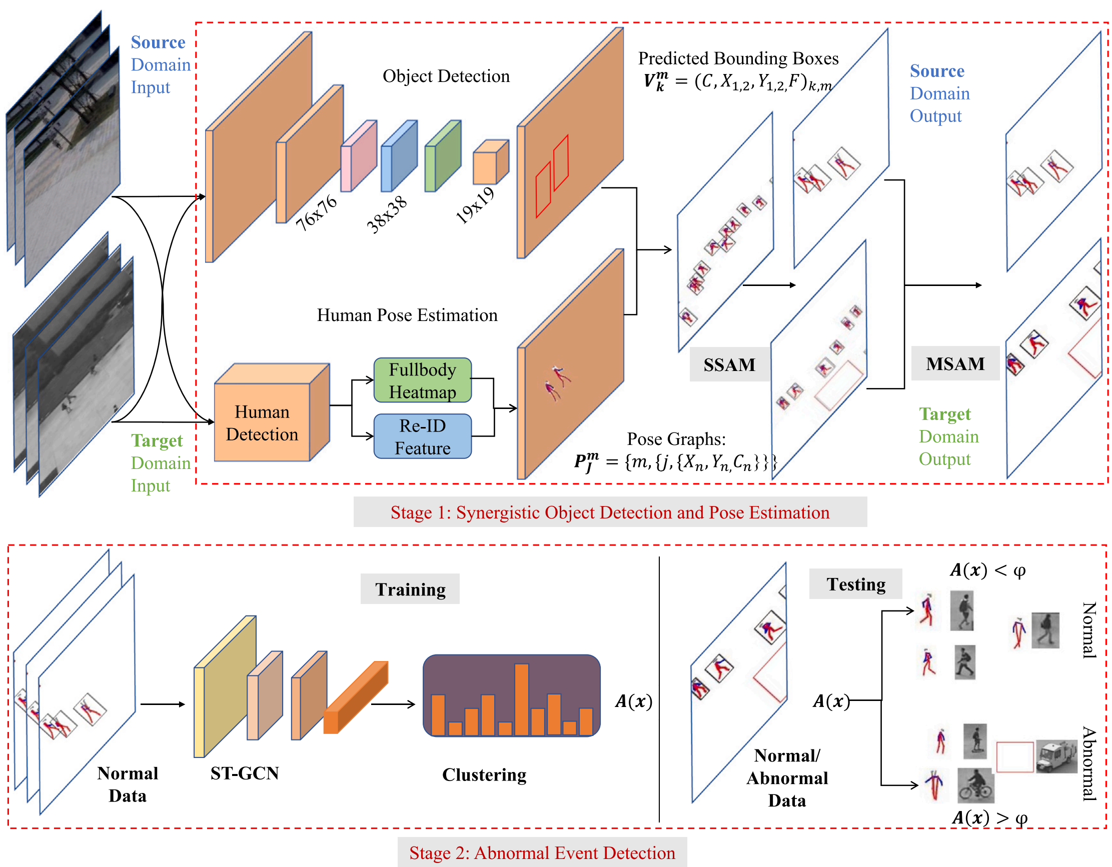

Fig. 3. The overall architecture of the proposed AED framework. Stage 1: Video sequences are clipped at the frame level and imported into object detection and pose estimation blocks. The extracted class and pose information are mapped in self scene-adaptive matching and mutual scene-adaptive matching to enhance pose estimation accuracy. Stage 2: In training, the improved pose data are fed into ST-GCAE networks to extract the latent vectors and further processed by a deep action-based clustering block. In testing, the videos are resized to adapt to the background size of training samples. The class and pose data are also extracted and mapped. The improved testing pose samples are put into the neural network to calculate normality scores at the frame level. Finally, the results of clustering and detection blocks are fused to do a binary classification to decide whether the frames are normal or abnormal.

2.2. Cross-scene aed and generalization

Cross-scene generalization, a crucial aspect of AED, refers to the capability of a model to apply its learning from one scene context to another diverse scene context without the necessity of additional training. This feature is particularly important considering the wide range of scenes a surveillance system encounters, from bustling urban intersections to quiet suburban parks. There have been several attempts to develop AED systems capable of cross-scene generalization. For instance, a cross-scene AED evaluation was discussed in [24–29]. The mainstream approach for scene-adaptive or scene-aware AED is to split the foreground and background, either with the aid of optical flow, where only moving objects are considered, or with the aid of pre-trained object detection networks to extract useful information.

For example, [25] uses YOLOv3 for object detection, followed by the computation of optical flow maps, which are processed by motion auto-encoders, while an appearance auto-encoder handles the object detections. The system employs adversarial training to distinguish between normal and pseudo-abnormal samples and integrates a segmentation branch to focus on foreground objects, ultimately using three binary classifiers to determine normality scores during inference. Meanwhile, the Ada-VAD framework [29] comprises a pretraining stage, where the model is trained to predict well on source normal samples while synthesizing abnormal samples to separate normal and abnormal embeddings. In the adaptation stage, mixup samples are generated by combining source and target samples, and the model is optimized to predict well on all types of samples, ensuring similar embeddings in an adversarial manner for effective adaptation to the target domain.

However, despite the apparent similarities between the scenes, the proposed cross-scene AED frameworks underperformed. One of the major reasons for the unsatisfactory performance was the significant intervention of background information and varying definitions of abnormal events across different datasets. Hence, these factors introduced a high level of complexity into the AED problem, rendering existing cross-scene generalization methods ineffective. Therefore, developing an AED framework that can effectively adapt across diverse scene contexts remains an open challenge and serves as a primary motivation for our work. Object detection results can be considered as prior knowledge to improve the performance of AED, which has been proven in many works [25,30,31]. As shown in Fig. 2, object detection can also extract foreground information from the environment. This improvement in cross-scene AED performance is possible as it reduces the interruption of background noise. However, object detection cannot completely remove the background, and some research shows that the background can sometimes be useful for AED.

2.3. Object detection and human pose estimation in cross- scene AED

Human pose estimation has increasingly become essential in developing advanced AED methods [32–34]. The unique advantage of human pose estimation lies in its ability to provide detailed and valuable information about human activities, which often form the basis of abnormal events in surveillance videos. By focusing on human poses, AED systems can detect and interpret normal and abnormal human behaviours more accurately. Several works have attempted to integrate human pose estimation into AED methods, considering it a robust scene-agnostic feature that can enhance cross-scene generalization [35]. The idea behind this approach is that human poses and movements remain relatively consistent across different scenes, making them a more reliable indicator of abnormal behaviours than scene-specific features.

Despite the potential advantages, incorporating human pose estimation into AED comes with challenges. These include the difficulty of accurately estimating poses in complex scenes, variations in human movements, and the influence of factors like occlusions and different camera perspectives [32]. Additionally, defining what constitutes an "abnormal" pose or sequence of poses can be complex, given the diversity of human behaviours. Despite these challenges, the integration of human pose estimation presents a promising direction for developing more sophisticated, scene-adaptive AED frameworks.

3. Proposed method

This paper presents a novel, scene-adaptive framework that addresses the challenges in the existing AED methods by integrating human pose estimation and object detection. Our approach pipeline is illustrated in Fig. 3, which consists of several key steps, each serving a unique function in our system's architecture.

In the initial stage, our system leverages synergistic pose estimation and object detection to incorporate context class results effectively. This improves the accuracy of the pose information and helps bridge the gap between different scenes. The system creates a series of pose sequences by effectively mapping the prior pose and class results. These pose sequences are further processed to generate more accurate and reliable data for subsequent stages. The generated pose sequences are then input into a spatial and temporal graph convolutional auto-encoder (ST-GCAE), a state-of-the-art model that excels at learning spatio-temporal feature representations. Based on these pose sequences, the ST-GCAE is trained to recognize normal human behaviours. It identifies patterns and relationships between the poses over time and space. A fully connected layer is appended to the ST-GCAE output to cluster human-related activities. This layer helps to group similar activities together and differentiate them from potentially abnormal events. In the final decision-making stage, the system makes a determination on the abnormality of events at the frame level. This is done based on the integrated normality scores from the ST-GCAE and the object detection module. An event is flagged as abnormal if it deviates significantly from the learned normal patterns.

3.1. Synergistic object detection and pose estimation

As shown in Fig. 4, The proposed work introduces a novel framework for AED that harmoniously combines object detection and pose estimation to improve the accuracy of surveillance video analysis.

Given a video sequence $\mathbf{V}$, each frame is initially processed using the YOLOv3 object detection model [36], a well-known and robust architecture for detecting and classifying objects in images. For each detected object in the ith frame, YOLOv3 outputs class labels, denoted as T, bounding box coordinates, denoted as $X_{1,2}$, $Y_{1,2}$, and confidence scores, denoted as C, respectively. If K objects are detected within a frame, the frame can be represented as a set of object detection tuples:

$$\mathbf{V}_K^m = [(T, X_{1,2}, Y_{1,2}, C)_{m,k}], k = 1, \dots, K \tag{1}$$

The class information obtained from object detection is crucial in initial target improvement and final AED steps, as it provides valuable context for interpreting scene activities.

Building on previous work on pose-level human tracking [35] and human action recognition [30,37], this framework also incorporates a pose-driven action recognition component. The crucial components of pose graphs are nodes representing physical locations and edges illustrating the vectors between two nodes, which are extracted from video sequences. Being a form of non-grid data, these pose graphs are estimated and fed into an ST-GCAE network [38]. The ST-GCAE captures latent vectors of normal actions, which subsequently aids in clustering abnormal actions in test videos.

Furthermore, pose information helps to mitigate the influences of varying background illumination and camera viewpoints across different datasets, thereby creating a conducive environment for robust cross-scene AED.

To extract pose graphs, this work employs the AlphaPose network [39], an accurate real-time multi-person pose estimator. It outputs pose graphs represented as a 3-D dictionary, which include pose identifications, frame numbers, 17 body landmarks, and the corresponding confidence scores. For the mth frame with J pose graphs detected, the output after pose estimation is given by:

$$\mathbf{P}_J^m = [m, j, X_n, Y_n, C_n], j = 1, \dots, J; n = 1, \dots, 17 \tag{2}$$

Overall, this integrated framework of object detection and pose estimation paves the way for more nuanced and precise AED in surveillance video data.

3.1.1. Self scene-adaptive matching

The proposed framework further incorporates a self scene-adaptive matching (SSAM) module designed to enhance the accuracy of pose information. This is achieved by integrating the results from object detection and pose estimation processes and applying stringent criteria to eliminate potential false alarms.

This framework considers two primary types of false alarms: **Wrong identification tracking**: This type of false alarm occurs when the difference between the current and previous frames' pose sequences exceeds a specified threshold. This discrepancy indicates that the pose has been incorrectly tracked between frames. In a crowded environment, if the appearances of multiple targets are similar, the pose estimator would also consider multiple targets into one pose sequence, Especially in the occlusion problem. The manifestations in the pose data are

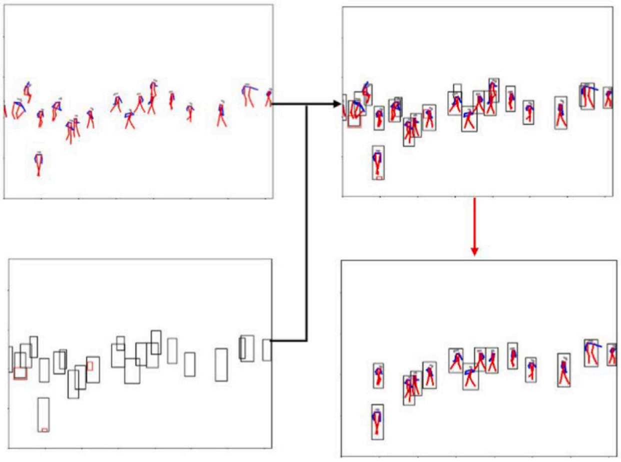

Fig. 4. A pose estimation sample (top left). The corresponding object detection sample (bottom left). The combined pose and bounding box information (top right). The enhanced pose graphs after mapping (bottom right).

that the pose localization between two short time frames is huge, and the frame numbers in one pose sequence are close; **Insufficient intersection**: If the intersection area between the estimated pose and the corresponding object bounding box is less than a certain threshold, it suggests a potential false alarm in either object detection or pose estimation.

Algorithm 1: Self Scene-Adaptive Matching

Input: Localization of $\mathbf{P}_J^m$: X_j^m, Y_j^m; The bounding box in S_k^m: $x_{1,k}^m$,$y_{2,k}^m$,$y_{1,k}^m$,$y_{2,k}^m$; The confidence: C_k^m.
Output: Trusted pose graph: $\mathbf{P}_T^m$

Initialize $t = 0$;
for $k = 0 \sim K$ **do**
    Compute the bounding boxes areas: $S_k^m = (y_{2,k}^m - y_{1,k}^m)(x_{2,k}^m - x_{1,k}^m)$
end
Compute the areas of each pose graph in $\mathbf{P}_J^m$: $P_j^m = |[max(Y_j^m) - min(Y_j^m)][max(X_j^m) - min(X_j^m)]|$ **for** $j = 0 \sim J$ **do**
    Compute the IoU scores of the pose graph: $IoU_{j,k}^m = \frac{P_j^m}{S_k^m}$
    if $IoU_{j,k}^m > \epsilon$ & $C_k^m > \tau$ **then**
        $P_t^m = P_j^m$
        $t = t + 1$
    end
end

3.1.2. Mutual scene-adaptive matching

This paragraph emphasizes the consideration of the spatial disparity between the source and target domains, leading us to propose a novel mutual scene-adaptive matching (MSAM) module. The primary objective of this approach is to ensure relative consistency in the sizes of training and testing poses, thereby improving the model's performance.

This algorithm comprises two principal approaches, both aimed at maintaining size consistency. The first approach involves resizing the test image to match the training image size. This adjustment effectively enlarges the test poses, bringing them in line with the size of the training poses. The second approach, on the other hand, focuses on adjusting the bounding box size of the initial pose in each video sequence to match the receptive field of the training poses. Subsequent poses in the sequence maintain the same ratio as the first pose, ensuring size uniformity across the sequence. Let S_{train} be the pose size in the training set, and S_{test} be the pose size in the testing set. Then, the transformation function for the first approach can be defined as:

$$S'_{test} = \alpha_1 \times \frac{S_{test}}{S_{train}} \tag{3}$$

where α_1 is the ratio of training and testing images' sizes and S'_{test} is the new size of the testing pose.

The second approach is represented as:

Let B_{1st} be the bounding box size of the first pose in a sequence and B_i be the bounding box size of the ith pose in the same sequence. We adjust the size of the bounding box for subsequent poses as follows:

$$B'_i = \alpha_2 \times \frac{B_i}{B_{1st}} \tag{4}$$

where α_2 is the ratio of the first pose graph's bounding box and test image sizes, and B'_i is the new bounding box size for the ith pose.

3.2. ST-GCAE and clustering implementation

The ST-GCAE and the deeply embedded clustering algorithm for refining pose information are introduced in Fig. 5. In this process, [38] is utilized to distil features corresponding to normal events. These features are then catalogued into an action dictionary through a step of deep-embedded clustering.

The procedure for feature extraction entails an analysis in the spatial domain, which includes the examination of the physical localization and relationships of body landmarks under a specific coordinate system. Concurrently, the temporal domain considers the changes in body landmarks over sequential frames. This process allows for the creation of the k_{th} pose graph in the i_{th} frames on video clips, represented as $\{N_k^i, E_k^i\}$.

Pose graph nodes: $N_k^i = \left\{\mu_{j,k}^i \in \mathbf{S}^D | j = 1, \ldots, 17\right\}$, delineate the localization of all landmarks. Here, $\mathbf{S}$ represents the set of all joints, while D is the dimensionality of these joints. On the other hand, $E_k^i = \{\epsilon_{j,k}^i\}$ symbolizes the edges or the connections between neighbouring landmarks. To facilitate the operation of the graph neural network, a fixed matrix A is established for all layers [40].

The formula governs the computation for this process:

$$G(N^i) = \Lambda^{-\frac{1}{2}}(A + I)\Lambda^{-\frac{1}{2}} N^i W^i \tag{5}$$

where N^i signifies the set of nodes, Λ refers to the degree matrix, I is the identity matrix denoting self-connections, and W^i represents the trainable weights in the network.

After extracting the spatio-temporal features via the neural network, these features are subjected to analysis and clustering. A clustering model is built, enabling the computation of the probability distribution of normal data.

During the training phase, the Kullback–Leibler (KL) divergence between the clustering probability distribution P and the distribution of the identified objects Q is minimized [32]. The loss function associated with the clustering step is expressed as:

$$L = KL(Q|P) = \sum_m \sum_n q_{m,n} log\left(\frac{q_{m,n}}{p_{m,n}}\right) \tag{6}$$

In this equation, m, n signifies the m_{th} pose attributed to the n_{th} cluster. This careful and systematic approach ensures a comprehensive understanding and utilization of the ST-GCAE model and the deep-embedded clustering algorithm.

3.3. Normality score calculation

During inference, anomaly scores are calculated with the results of pose normality scores N_p from the graph neural network and the class normality scores from object detection and represented as:

$$N_t = \delta \frac{\sum_{k=1}^K {N_{pt}}^k}{K} + (1 - \delta) \frac{\sum_{k=1}^K R_t^k C_t^k W_i}{U} \tag{7}$$

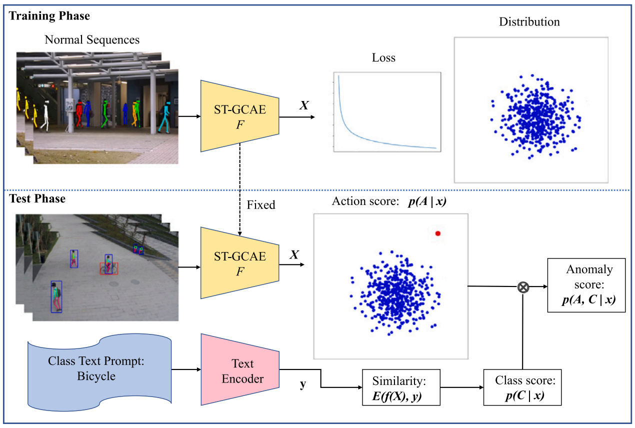

Fig. 5. The schematic diagram of ST-GCAE and Clustering implementation.

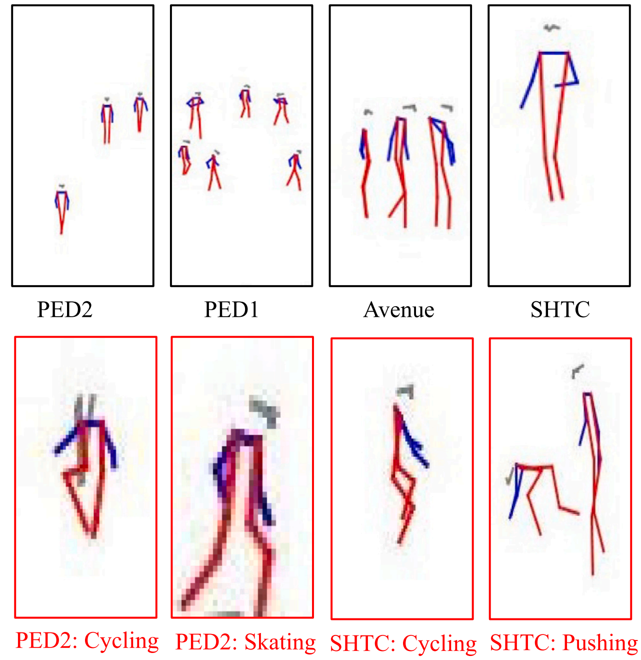

Fig. 6. The pose-level data samples are presented. The top row images depict the normal poses from the training part of the datasets, while the bottom row images represent the poses associated with abnormal events. The pose size has been normalized for better viewing.

where δ is the weight for pose information, R is the area of bounding boxes, C is the corresponding confidence, W is the weights of objects, and U is the frame size.

4. Experiments

In this section, we first introduce our datasets, evaluation metrics, and experimental settings. Then, we describe the quantitative results in both cross-scene and cross-database domains. The proposed framework is trained in one-scene video sequences of the SHTC dataset and tested on the rest of the video sequences from the same dataset. Table 1 details the AUC performance on different scenes and compares it with two related works for AED on the SHTC dataset. Meanwhile, we present our experimental efficiency and the results of the ablation study. Finally, we provide a comparison with state-of-the-art AED methods.

4.1. Datasets

Our work aims to demonstrate the advantages of the pose-level AED algorithm compared to image-based methods when detecting unseen scenes. Thus, in addition to utilizing existing benchmark datasets, we conduct training on video clips in a laboratory environment, as described in [30]. This experiment has five training sets: UCSD PED1, UCSD PED2, CUHK Avenue, ShanghaiTech Campus, and BMbD. Meanwhile, the proposed methods are tested on the test set portion of two specific datasets: UCSD PED2, which is a small-scale AED dataset containing only grey images and a small number of abnormal events in a fixed background, and ShanghaiTech Campus, which is a large-scale AED dataset with RGB images across 13 different scenes. Fig. 6 shows some representative examples from the training and test sets. Each dataset is introduced and listed below.

UCSD PED [41]: This dataset contains two parts: UCSD PED1 and PED2. PED1 consists of 34 training videos and 36 testing videos with a frame resolution of 238×158, while PED2 consists of 16 training videos and 12 testing videos with a frame resolution of 360×240. The datasets are recorded from fixed viewpoints, ensuring that the scene context of the training and testing sets is consistent.

ShanghaiTech Campus [23] (SHTC): This dataset is one of the most challenging datasets for AED. It comprises 330 training video sequences and 107 testing video sequences, with complex illuminations and 13 different scenes, each having a frame resolution of 856×480.

CUHK Avenue [42]: This dataset contains 16 training video clips and 21 testing video clips with a consistent background. However, most abnormal events are unusual behaviours such as throwing, loitering, and fast-running.

4.2. Performance measures and parameters

The underlying assumption is that the test data only contain positive and negative classes, with the predicted results being true or false. This delineation generates four possible outcomes: true positive (TP), true negative (TN), false positive (FP), and false negative (FN). The true positive rate (TPR) and false positive rate (FPR) are defined by the following equations:

$$TPR = \frac{TP}{P} = \frac{TP}{TP + FN} \tag{8}$$

$$FPR = \frac{FP}{N} = \frac{FP}{FP + TN} \tag{9}$$

where TP and P represent the true positive and actual positive (abnormal) samples in video sequences, respectively; FN is the false negative; FP and N represent the false positive and actual negative (normal) samples, respectively; and TN is the true negative in video sequences. The receiver operating characteristic (ROC) curve is plotted with the TPR on the y-axis and FPR on the x-axis. The area under the ROC curve represents the Area Under the Curve (AUC), a scalar representing the area under the ROC curve.

4.3. Cross-scene quantitative results

In the realm of AUC, two interpretations, Micro-AUC and Macro-AUC, are frequently discussed [25,43,44]. Their importance is amplified in contexts where datasets are imbalanced, a common occurrence in AED due to the inherent rarity of such events. Micro-AUC aggregates contributions from all classes to yield a unified metric. This is especially pertinent when evaluating a model's performance across all instances, even if some categories are underrepresented. High Micro-AUC scores often denote robust all-around model performance.

Conversely, Macro-AUC calculates the metric independently for each category, followed by averaging. This ensures every category, regardless of its representation, is equally weighted. Thus, a high Macro-AUC implies adeptness in detecting varied anomalies, even those

Table 1
AUC performance on the SHTC dataset when training in the one-camera scene and testing in the rest of the unseen scenes.

Method	Cam-1	Cam-2	Cam-3	Cam-4	Cam-5	Cam-6	Cam-7	Cam-8	Cam-9	Cam-10	Cam-11	Cam-12
Frame-pred [23]	0.678	0.618	0.663	0.659	0.698	0.735	0.681	0.619	0.674	0.679	0.655	0.651
Doshi et al. [15]	0.753	0.707	0.761	0.681	0.784	0.814	0.789	0.626	0.706	0.663	0.753	0.719
w/o SMSAM	**0.792**	**0.826**	**0.839**	**0.830**	**0.824**	0.792	0.782	**0.763**	**0.787**	**0.749**	**0.779**	**0.738**
SMSAM	**0.807**	**0.831**	**0.817**	**0.832**	**0.835**	**0.817**	**0.826**	**0.805**	**0.826**	**0.790**	**0.823**	**0.831**

Table 2
AUC performance on the SHTC dataset when training in 3,5,10 scenarios and testing in the 8th, 10th, and 12th unseen scenarios.

Training scenarios	Training video sequences	Training poses	AUC
Cam-1 - Cam-3	104	255 498	0.823
Cam-1 - Cam-5	190	574 469	0.842
Cam-1 - Cam-7, Cam-9, Cam-11	255	714 513	0.826

sparsely represented. Given the nature of surveillance video analysis, datasets often lean towards normal events due to the infrequency of abnormal ones. This imbalance accentuates the relevance of the AUC metric [45]. While Micro-AUC's importance is paramount in environments like security, where every anomaly's detection is crucial, Macro-AUC becomes indispensable when anomalies span a spectrum of types.

In summary, both Micro and Macro AUC offer comprehensive insight into a model's performance in AED. Where Micro-AUC offers a deep dive into prediction accuracy, Macro-AUC ensures no abnormality, however rare, goes unnoticed.

We utilize ST-GCAE [38] as our backbone architecture and a deeply embedded clustering [46] for the action classification. The learning rate parameter for ST-GCAE is 0.0001. The dropout training parameter is 0.3. The patch size for training is 16. And the activation function used is $ReLU$. The clustering parameter and batch size are defaulted as 10 and 512. The PyTorch framework implements the experiments on four GeForce GTX 1080Ti GPUs.

Table 1 presents the AUC performance of different methods on the SHTC dataset across multiple camera scenes. 'Cam-1' means training in the first scenes and testing in the rest of the unseen scenes. The methods analysed include Liu et al. [23] and Doshi et al. [15], which are conventional image-level AED methods, and two variants of our proposed pose-level methods, labelled as 'w/o SMSAM' (without self and mutual scene-adaptive matching) and 'SMSAM' (self and mutual scene-adaptive matching). The AUC performance shows that the proposed method, both with and without SMSAM, generally outperforms the baseline models proposed by Liu et al. [23] and Doshi et al. [15]. In particular, the highest AUC performance across all camera scenes is often achieved by the proposed SMSAM method, emphasizing the significance of self and mutual scene-adaptive matching in the efficacy of our model. It is also noteworthy that the proposed w/o SMSAM method, even without the self and mutual scene-adaptive matching components, can still yield superior results compared to the baseline models in most camera scenes, which indicates that pose-level AED methods are more capable than image-level AED methods for the cross-scene AED task. This further validates the robustness and effectiveness of the core components of our proposed methods. While Doshi et al. [15] also achieves higher AUC performance compared to Liu et al. [23], it is worth highlighting that the proposed method exhibits a more substantial improvement, suggesting that our methodology addresses some of the limitations of these baseline models.

Table 2 shows the AUC results when training in 3, 5, and 10 scenarios and testing in three other unseen scenarios. The AUC performance is stable overall. It can also be observed that the AUC performance improves when training increases from 3 scenarios to 5 scenarios. However, the AUC performance decreases when training increases from 5 scenarios to 10 scenarios, possibly due to over-fitting, which decreases the AED's effectiveness.

These results affirm our proposed method's robustness and superior performance in handling different camera scenes for AED in video sequences. Future work could explore enhancements to the self scene-adaptive matching and mutual scene-adaptive matching components, as they have demonstrated potential for further improving model performance.

4.4. Cross-database quantitative results

Table 3 showcases the AUC performance comparison in the same dataset and across different datasets to evaluate scene adaptability. It contrasts 7 distinct cross-scene AED algorithms: Recons-based [47], Frame-pred [23], HF-VAD [27], Ada-VAD [28], Jigsaw [29], Few-show [24], and Adversarial Training [25]. Our proposed pose-level method is further divided into three subcategories: SSAM, MSAM, and SMSAM. The evaluation was conducted across two primary testing environments, PED2 and SHTC, with training carried out on various datasets, including PED2, PED1, SHTC, and Avenue. Our proposed method performs consistently during all testing scenarios. Particularly in the scenarios with training on PED1, testing on PED2, and training on PED2 and testing on SHTC, our proposed SMSAM approach demonstrates superior adaptability with the highest AUC scores of 0.957 and 0.833, respectively. Overall, Table 3 highlights the robustness and efficacy of our proposed pose-level algorithms in scene adaptability, outperforming traditional cross-scene AED methods.

Table 3 also details the performance of our models for same-scene AED on the UCSD and SHTC datasets. The AUC performance decreases on cross-scene evaluation compared with same-scene evaluation. In our proposed framework, the performance of training on the SHTC dataset and testing on the PED2 dataset slightly drops. On the contrary, the AUC performance drops dramatically. Table 3 also demonstrates the advantages of the pose-based model for scene-adaptive AED, as pose-level action classification models are robust for cross-scene evaluation with the combination of object detection. Compared with other image-based AED work results, the proposed framework is more suitable for scene-adaptive AED tasks.

4.5. Model complexity and inference speed

To evaluate the detection efficiency of the proposed AED method, a comparison was conducted with three baseline AED methods: rGAN [48], Frame-pred [23], and MPN [49]. The rGAN and MPN methods are based on meta-learning, and the Frame-pred method is based on future frame prediction, all of which are efficient for training and testing in AED. The evaluation was performed across three dimensions: model parameter count, execution time, and frames per second (FPS).

The results in Table 4 of the parameter count analysis demonstrate that the proposed frameworks, driven by pose information, exhibit a notable advantage in terms of model complexity and resource utilization. With a significantly lower parameter count than the baseline models, the proposed method offers potential benefits in efficiency and resource allocation. Considering execution time, the comparison reveals that the rGAN and Frame-pred models require relatively longer execution times, while the MPN and our proposed models demonstrate significantly faster processing. In this context, our proposed framework achieves the best results, with an execution time of 4 s, suggesting that

Table 3
AUC performance on different datasets for scene adaptability.

Train⟶ Test	Recons -based [47]	Frame -pred [23]	HF -VAD [27]	Ada- VAD [28]	Jig- saw [29]	Few- shot [24]	Adversarial training [25]		**Proposed**			
							Micro	Macro	SSAM	MSAM	**SMSAM** Micro	Macro
PED2⟶ PED2	0.874	0.954	–	–	–	0.968	**0.987**	**0.997**	0.944	–	0.944	0.969
PED1⟶ PED2	0.768	0.898	–	–	–	–	–	–	0.908	0.894	0.921	**0.957**
SHTC⟶ PED2	0.791	0.895	0.855	**0.975**	0.952	0.940	0.906	0.957	0.938	0.916	0.945	**0.966**
Avenue⟶ PED2	0.689	0.870	0.908	**0.984**	0.820	–	0.870	0.972	0.877	0.896	**0.909**	0.897
SHTC⟶ SHTC	0.650	0.729	–	–	–	0.752	0.827	**0.893**	0.839	–	**0.839**	0.846
Avenue⟶ SHTC	0.572	0.738	0.764	0.763	0.794	–	0.768	**0.863**	0.796	0.784	**0.818**	0.841
PED2⟶ SHTC	0.575	0.717	0.749	0.750	0.689	–	–	–	0.785	0.743	0.794	**0.833**
PED1⟶ SHTC	0.545	0.714	–	–	–	–	–	–	0.738	0.719	0.772	**0.801**

Table 4
Comparison in detection efficiency with image-level and pose-level AED methods.

	Model parameter (↓) (millions)	Execution time (↓) (s)	FPS (↑)
rGAN [48]	19.0	957	2.1
Frame-pred [23]	88.2	1568	2.9
MPN [49]	12.7	12	166.8
Proposed	**0.8**	**4**	**515.4**

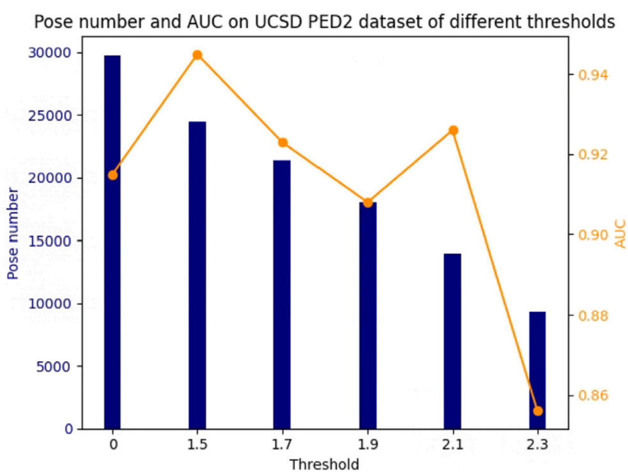

Fig. 7. The different thresholds setting for SSAM.

it balances computational efficiency and detection accuracy. Furthermore, evaluation in terms of frames per second (FPS) provides insights into the models' capabilities. Our proposed framework achieves an FPS of 515.4, indicating its ability to process video frames relatively faster than rGAN, Frame-pred, and MPN. It demonstrates promising detection performance while maintaining reasonable computational efficiency.

These findings highlight the advantages of pose-level methods in model complexity, execution time, and processing performance for AED. Our proposed framework shows promising parameter efficiency and processing capabilities results, primarily because it is a pose-oriented model. Further investigations and optimizations can be pursued as future work further to enhance the overall detection accuracy of the proposed algorithms.

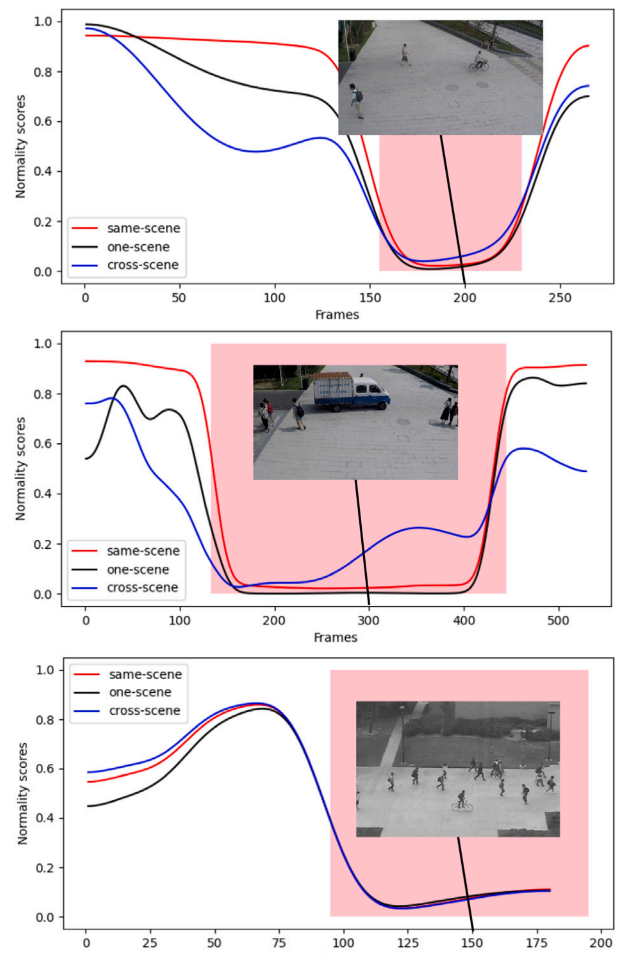

Fig. 8. Normality scores were evaluated using different scene training methods. The same-scene method involves training and testing on the same dataset. The one-scene method involves training on one scene of a dataset and testing on the unseen scenes of the same dataset. The cross-scene method involves training on one dataset and testing on another dataset. The specific train and test results are as follows. (Top (cycling case) & Middle (driving case)): Same-scene from SHTC to SHTC; One-scene from SHTC to the rest of SHTC; and Cross-scene from PED2 to SHTC. (Bottom): Same-scene from PED2 to PED2; One-scene from SHTC to PED2; and Cross-scene: from SHTC to PED2.

4.6. Ablation study

Fig. 7 shows the number of poses and AUC performance on the UCSD PED2 dataset under different thresholds. To ensure high-quality pose information, minimizing the impact of incorrect identifications, tracking errors, and insufficient intersections is crucial. The threshold process removes estimated pose graphs with low confidence scores or short sequence lengths. By applying this threshold, we can filter out unreliable pose estimates, leading to a noticeable improvement in AUC performance. This step highlights the importance of confidence and sequence length in maintaining the accuracy and reliability of pose-based abnormal event detection.

Fig. 8 visualizes the distribution of normality scores on the SHTC and PED2 datasets for the same-scene, one-scene, and cross-scene methods, revealing critical insights into the performance and characteristics

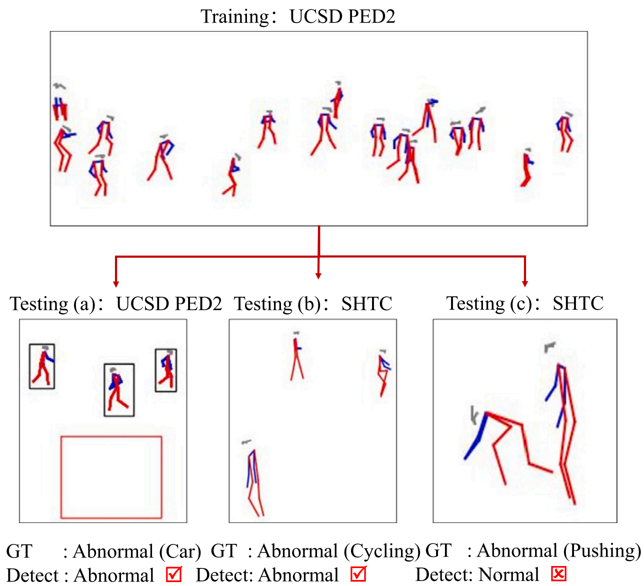

Fig. 9. Case study for AED.

Table 5
AUC performance for AED compared with state-of-the-art methods.

	SHTC	PED2	PED1
Conv-AE [18]	0.609	0.811	0.750
Stacked RNN [50]	0.680	0.922	–
ConvLSTM-AE [51]	–	0.881	0.755
Zhao et al. [52]	–	0.912	0.923
Frame-pred [23]	0.728	0.954	0.831
AnoPCN [53]	0.736	0.968	–
Morais et al. [54]	0.734	–	–
Zhou et al. [7]	–	0.960	0.839
Markovitz1 et al. [32]	0.761	–	–
Dual-GAN [55]	0.737	0.956	–
Yang et al. [16]	–	0.940	–
Hyun et al. [56]	0.740	0.972	–
MPN [49]	0.738	0.969	–
Zhang et al. [20]	0.803	0.929	**0.942**
Zhong et al. [57]	0.745	0.981	–
Wang et al. [58]	0.754	0.965	–
ROADMAP [59]	0.766	0.963	–
AMS-AE [60]	0.772	**0.996**	–
STR-VAD [61]	0.732	0.984	–
Few-shot [24]	0.752	0.968	–
Adversarial training [25]	0.827	0.987	–
Recons-based [47]	0.650	0.874	0.899
Proposed (same-scene)	**0.839**	0.944	0.748
Proposed (cross-scene)	0.818	0.945	0.726

of each technique. The sub-figures present the normality scores for abnormal actions, specifically cycling (Top) and driving (Middle), visually comparing the different approaches. In general, all three methods can separate abnormal frames from normal samples. However, from the top and middle two subfigures, which are training on SHTC, one scene of SHTC, and PED2, and then testing on SHTC, the same-scene training approach naturally exhibits the most accurate detection, as it has direct access to specific scene information and can fine-tune the model. This approach provides a more nuanced understanding of the scene context, allowing for more precise normality scores.

The bottom subfigure shows some differences: training on PED2, one scene of SHTC, and SHTC, and then testing on PED2. Though all three methods perform similarly when detecting anomalies, the performance of the cross-scene technique is slightly better than the other two techniques. This is because the training model is based on a large-scale dataset. One-scene training can be limited by the lack of diversity in training data, which might result in reduced generalization ability. Conversely, cross-scene training demonstrates the potential for more versatile and flexible AED, as it adapts to new scenes with relative success.

These results underline the importance of context and scene-specific information in AED. The comparative study of same-scene, one-scene, and cross-scene training sheds light on the trade-offs between accuracy, efficiency, and adaptability. While same-scene training may yield the best performance for specific applications, the cross-scene approach offers a promising direction for more generalized and scalable solutions. The findings also suggest areas for future research, such as developing more robust cross-scene algorithms that can maintain high performance without sacrificing adaptability across various scene contexts.

4.7. Case study

Fig. 9 shows three cases of testing samples processed with the training step in the UCSD PED2 dataset. Testing (a) highlights the limitation of pose-oriented methods: the algorithm can only detect human-related AED. The car should not be detected initially since the driver is occluded and does not generate pose information. However, with object detection, the car can be correctly identified. Testing (b) and Testing (c) should be considered together. Since cycling is both a behaviour-catalog and object-catalog abnormal event, it is easier to detect it using a synergistic object detection and pose estimation system. However, pushing or fighting is an abnormal behaviour that is hard to detect due to the small changes between surrounding frames, and the behaviour is similar to walking. Thus, the definition of anomaly is the main task of the next step.

4.8. Comparison with state-of-the-art

Table 5 compares the AUC performance between the proposed scene-adaptive framework and other state-of-the-art methods. The proposed approach achieves the highest AUC score, 0.839, on the SHTC dataset and a relatively high AUC score, 0.944 on the UCSD PED2 dataset. Existing methods' comparison in Table 5 is only valid for same-scene scenarios, and the methods are mainly reconstruction-based models such as ConvAE [18], and AMS-AE [60] and predictive models such as Frame-pred [23]. Compared with the state-of-the-art results, our proposed framework outperformed the SHTC dataset and has a comparable performance on the PED1 and PED2 datasets. The AUC performance of our proposed model on the PED1 dataset is relatively low. This is because the PED1 dataset contains mainly object-related abnormal events, lacks action-related abnormal events, and has the limitation of low resolution. Future work can address this issue by including motion features. Our pose-oriented cross-scene adaptive AED framework suits human-related unseen data scenarios. The proposed synergistic pose estimation and object detection strategy effectively integrate the pose and class information for feature extraction in the challenging cross-data AED tasks. Moreover, the AUC values of our proposed methods remain stable under different training settings (same-scene and cross-scene), which is another advantage of the pose-level scene-adaptive AED.

5. Conclusions

In surveillance videos, we presented a pose-oriented self and mutual scene-adaptive matching method for AED. With the synergistic pose estimation and object detection, pose data is extracted and enhanced to feed into ST-GCAE for action classification and a downstream AED task. The extensive evaluations on the SHTC and UCSD PED2 datasets demonstrated the efficacy of the proposed pose-level models, which are further accentuated by their exceptional scene adaptability. The introduction of the SMSAM component was important in achieving superior results across diverse camera scenes, suggesting the necessity of scene-adaptive components for cross-scene anomaly detection. Furthermore, our experiments regarding model complexity and inference speed highlighted the efficiency of the pose-level approach. With a significant reduction in model parameters compared to traditional models and achieving faster processing speeds, the proposed methods ensure real-time anomaly detection without compromising accuracy. Besides, compared with the recent work, our proposed pose-level method achieved relatively high AUC scores,0.839 and 0.944, on the SHTC and UCSD PED2 datasets. This suggests that the devised method adeptly navigates the challenges inherent in AED across various contexts.

The solution of the novel cross-scene training for AED has huge potential in real-world applications. The results of pose-level cross-scene training for AED are feasible on different datasets with the same tasks, similar activities, and different backgrounds. Compared with image-level cross-scene training for AED, the scene-agnostic pose information is more effective for the cross-scene AED task. Through the results of cross-scene training decrease, the AUC performances of training on complicated datasets and testing on simple datasets are acceptable.

CRediT authorship contribution statement

Yuxing Yang: Writing – original draft, Validation, Software, Methodology. **Leiyu Xie:** Validation, Methodology, Investigation. **Zeyu Fu:** Writing – review & editing, Methodology. **Jiawei Yan:** Writing – review & editing. **Syed Mohsen Naqvi:** Writing – review & editing, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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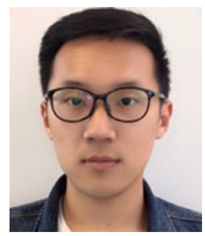

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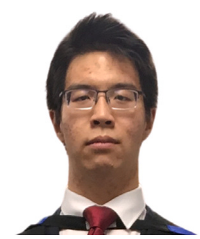

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